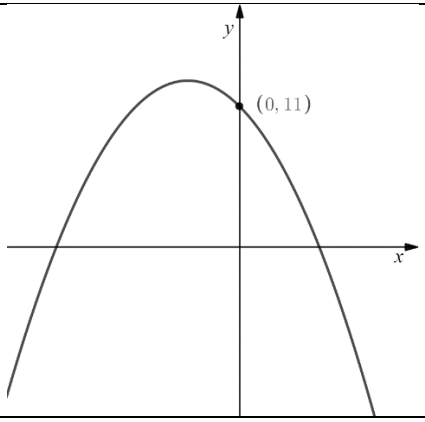
1.	$\int\left(\frac{8x^3}{5}-\frac{2}{3x^4}-1\right)\mathrm{d}x=\frac{1}{4}\times\frac{8x^4}{5}-\frac{2}{3}\times\frac{1}{-3}x^{-3}-x$	M1 A1
	$\frac{2}{5}x^4+\frac{2}{9}x^{-3}-x+c$	A1 A1
		(4)
		Total 4

2(a)	$f(x) = 11 - 4x - 2x^2$ $\Rightarrow ... \underline{-2}(2x + x^2) ...$ or $\Rightarrow ... \underline{-2}(2x + x^2 ...)$	B1
	$...(2x + x^2) \Rightarrow ....((x+1)^2 \pm ...)$	M1
	$(f(x) =)\ 13 - 2(x+1)^2$	A1
		(3)
(b)		M1
		A1
		(2)
(c)	$x = -1$	B1ft
		(1)
		Total 6
Alt(a)	$a + b(x^2 + 2cx + c^2) = 11 - 4x - 2x^2$ $b = -2$	B1
	$2bc = -4 \Rightarrow c = ... (= 1)$	M1
	$a + bc^2 = -4 \Rightarrow a = ... (= 13)$ $(f(x) =)\ 13 - 2(x+1)^2$	A1

3(i)	$\sqrt{8} = 2\sqrt{2}$ seen anywhere in the solution	B1
	$\left(x+\sqrt{2}\right)^2 + \left(3x-5\sqrt{8}\right)^2$ $= x^2 + 2x\sqrt{2} + 2 + 9x^2 - 30x\sqrt{8} + 25\times 8$	M1
	$= 10x^2 - 58x\sqrt{2} + 202$	A1
		(3)
(ii)	$\sqrt{3}\left(4y-3\sqrt{3}\right) = 5y+\sqrt{3}$ $\Rightarrow 4\sqrt{3}y - 9 = 5y + \sqrt{3}$ $\Rightarrow 4\sqrt{3}y - 5y = 9 + \sqrt{3}$	M1
	$\Rightarrow y = ...$ or $\Rightarrow ky = ...$ eg $y = "\dfrac{9+\sqrt{3}}{4\sqrt{3}-5}"$ or $"23"y = "9+\sqrt{3}"$	dM1
	$y = \dfrac{9+\sqrt{3}}{4\sqrt{3}-5} \times \dfrac{4\sqrt{3}+5}{4\sqrt{3}+5} \Rightarrow y = \dfrac{...}{"23"}$	ddM1
	$y = \dfrac{57}{23} + \dfrac{41}{23}\sqrt{3}$ (or $y = 2\dfrac{11}{23} + 1\dfrac{18}{23}\sqrt{3}$)	A1
		(4)
	(ii) Alternative 1:	
	$\sqrt{3}\left(4p + 4q\sqrt{3} - 3\sqrt{3}\right) = 5\left(p+q\sqrt{3}\right) + \sqrt{3}$ $\Rightarrow 4p\sqrt{3} + 12q - 9 = 5p + 5q\sqrt{3} + \sqrt{3}$ $\Rightarrow 4p = 5q+1,\ 12q - 9 = 5p$	M1 dM1
	$\Rightarrow 4p = 5q+1,\ 12q - 9 = 5p$ $\Rightarrow p = ...,\ q = ...$	ddM1
	$y = \dfrac{57}{23} + \dfrac{41}{23}\sqrt{3}$ (or $y = 2\dfrac{11}{23} + 1\dfrac{18}{23}\sqrt{3}$)	A1
		Total 7

4(a)	$x+y=6,\ y=6x-2x^2+1$ $\Rightarrow 6-x=6x-2x^2+1$ $\Rightarrow 2x^2-7x+5=0$ oe	M1
	$2x^2-7x+5=0 \Rightarrow (2x-5)(x-1)=0$ $\Rightarrow x=\frac{5}{2},\ 1$	M1
	$x="\frac{5}{2}" \Rightarrow y=\frac{7}{2}$ or $x="1" \Rightarrow y=5$	dM1
	$(1,\ 5)$ and $(2.5,\ 3.5)$	A1
		(4)
(b)	$y \geqslant 6x-2x^2+1$ oe	M1
	$x+y \leqslant 6$ oe $x \geqslant a$ where $1 \leqslant a \leqslant 2.5$	A1
	$(\text{or } a \leqslant x \leqslant b \text{ where } 1 \leqslant a \leqslant 2.5,\ b \geqslant 6)$ $y \geqslant 0$ (or $0 \leqslant y \leqslant c$ where $c \geqslant 3.5$)	A1
		(3)
		Total 7

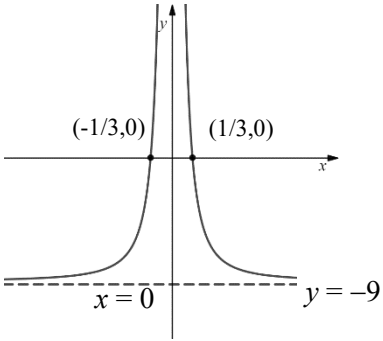
5(a)	$\angle BOD = \pi - 2\times 0.7 = 1.742$*	B1*
		(1)
(b)	Area of $BOD = \frac{1}{2}\times 3^2 \sin 1.742$ $(=$ awrt $4.43)$	M1
	Area of R is: $\frac{1}{2}\times 3^2 \times 1.742 - \frac{1}{2}\times 3^2 \sin 1.742$ or $\frac{1}{2}\times \pi \times 3^2 - \frac{1}{2}\times 3^2 \sin 1.742 - 2\times \frac{1}{2}\times 3^2 \times 0.7$	dM1
	= awrt 3.4 (m^2)	A1
		(3)
(c)	$BD = \sqrt{3^2+3^2-2\times 3\times 3\cos 1.742}$ $(=$ awrt $4.59)$ or $BD = 2\times 3\sin\left(\frac{1.742}{2}\right)$ or $BD = 2\times 3\cos 0.7$ or $BD = \frac{3\sin 1.742}{\sin\left(\frac{\pi - 1.742}{2}\right)}$ or arc $BCD = 3\times 1.742$ (= 5.226)	M1
	Perimeter of R is: $3\times 1.742 + "BD"$	dM1
	= awrt 9.8 (m)	A1
		(3)
		Total 7

6(a)(i)	$x=4 \Rightarrow \mathrm{f}'(4)=\dfrac{(4+3)^2}{4\sqrt{4}}=\dfrac{49}{8}$ (6.125)	B1
(ii)	$y-20="\dfrac{49}{8}"(x-4)$	M1A1ft
	$49x-8y-36=0$	A1
		(4)
(b)	$\mathrm{f}'(x)=\dfrac{(x+3)^2}{x\sqrt{x}}=\dfrac{x^2+6x+9}{x\sqrt{x}}$ $=\dfrac{x^2}{x\sqrt{x}}+\dfrac{6x}{x\sqrt{x}}+\dfrac{9}{x\sqrt{x}}=\ldots$	M1
	$=x^{\frac{1}{2}}+6x^{-\frac{1}{2}}+9x^{-\frac{3}{2}}$	A1
	$\left(\mathrm{f}(x)=\right)\dfrac{2}{3}x^{\frac{3}{2}}+12x^{\frac{1}{2}}-18x^{-\frac{1}{2}}+c$	dM1 A1A1
	$20=\dfrac{2}{3}(4)^{\frac{3}{2}}+12(4)^{\frac{1}{2}}-18(4)^{-\frac{1}{2}}+c \Rightarrow c=\ldots$	M1
	$\left(\mathrm{f}(x)=\right)\dfrac{2}{3}x^{\frac{3}{2}}+12x^{\frac{1}{2}}-18x^{-\frac{1}{2}}-\dfrac{1}{3}$	A1
		(7)
		Total 11

7(a)	$4\times -2\times -9=72$ $p=\text{"}72\text{"}-50$	M1
	$(p=)\ 22$	A1
		(2)
(b)	$(q=)-4,\ 2,\ 4.5$	B1B1
		(2)
(c)	$\mathrm{f}(x)=(x+4)(x-2)(2x-9)$ $\mathrm{f}(x)=(x^2+2x-8)(2x-9)=...x^3\pm...x^2\pm...x\ (\pm...)$	M1
	$=2x^3-5x^2-34x\quad(+72)$	A1
	$(\mathrm{f}'(x)=)\ 6x^2-10x-34$	M1A1
		(4)
(d)	$\text{"}6x^2-10x-34\text{"}=-18$ $\text{"}6x^2-10x-16\text{"}=0\Rightarrow x=...\left(-1,\ \frac{8}{3}\right)$	M1
	$-1<x<\frac{8}{3}$	dM1A1
		(3)
		Total 11

8(a)	$\frac{2}{5}$ or decimal equivalent	B1
		(1)
(b)	$m_N = -1 \div "\frac{2}{5}"$	M1
	$y + 2 = "-\frac{5}{2}"(x - 6)$	M1
	$y = -\frac{5}{2}x + 13$	A1
		(3)
(c)	$"-\frac{5}{2}x + 13" = "\frac{2}{5}x + \frac{7}{5}" \Rightarrow "\frac{29}{10}x" = "\frac{58}{5}" \Rightarrow x = ...(= 4)$ or $"\frac{5}{2}y - \frac{7}{2}" = "-\frac{2}{5}y + \frac{26}{5}" \Rightarrow "\frac{29}{10}y" = "\frac{87}{10}" \Rightarrow y = ...(= 3)$	M1
	$x = "4" \Rightarrow y = ...$ or $y = "3" \Rightarrow x = ...$	dM1
	$(4, 3)$	A1
		(3)
(d)	$(2, 8)$	B1B1
		(2)
		Total 9

9(a)	$(A=)-3$	B1
		(1)
(b)	$y=3$	B1
	eg $x=30+5\times180$ or $x=210+720$ or $x=180+2\times360+30$	M1
	$x=930$	A1
		(3)
		Total 4

10(a)		B1B1B1B1
		(4)
(b)	$k < 0$	B1
	$\frac{1}{x^2} - 9 = kx^2 \Rightarrow 1 - 9x^2 = kx^4$	M1
	$1 - 9x^2 = kx^4 \Rightarrow kx^4 + 9x^2 - 1 = 0$ Require $b^2 - 4ac \Rightarrow 81 - 4 \times k \times -1$	M1
	Critical value $\left(k =\right) -\frac{81}{4}$	A1
	$-\frac{81}{4} < k < 0$	A1cso
		(5)
		Total 9